

Lecture Slide – 11

Interpolations

INTERPOLATION

Interpolation is a **numerical method** used to estimate the value of a function for an intermediate point **within** the range of a discrete set of known data points. A standard idea in interpolation is to find a polynomial $p_n(x)$ of degree n that assumes the given $n+1$ values; thus

$$p_n(x_0) = f_0, \quad p_n(x_1) = f_1, \quad \cdots, \quad p_n(x_n) = f_n.$$

We call this $p_n(x)$ an interpolation polynomial and $x_0, \cdots, x_n$ the nodes. And if $f(x)$ is a mathematical function, we call $p_n(x)$ a polynomial approximation of f .

We use $p_n(x)$ to get the value of f for a specific x between x_0 to x_n (“interpolation”) or sometimes outside this interval (“extrapolation”).

Types of INTERPOLATION

There are several interpolation techniques:

Linear Interpolation – Approximates between two data points using a straight line.

Polynomial Interpolation – Fits an n th-degree polynomial through $n+1$ data points. *The general goal of fitting a polynomial through data points.*

Lagrange Interpolation – Uses a set of basis polynomials to construct the interpolating polynomial explicitly. *A specific method to achieve that goal using explicit basis polynomials.*

Newton's Divided Difference Interpolation – Builds the polynomial incrementally using divided differences.

Spline Interpolation – Fits low-degree polynomials piecewise to ensure smoothness.

Direct Method

Given ' $n+1$ ' data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, pass a polynomial of order ' n ' through the data as given below:

$$y = a_0 + a_1x + \dots + a_nx^n.$$

where $a_0, a_1, \dots, a_n$ are real constants.

- Set up ' $n+1$ ' equations to find ' $n+1$ ' constants.
- To find the value ' y ' at a given value of ' x ', simply substitute the value of ' x ' in the above polynomial.

Example 1

The upward velocity of a rocket is given as a function of time in Table 1.

Find the velocity at $t=16$ seconds using the direct method for linear interpolation.

Table 1: Velocity as a function of time.



$t, (s)$	$v(t), (m/s)$
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

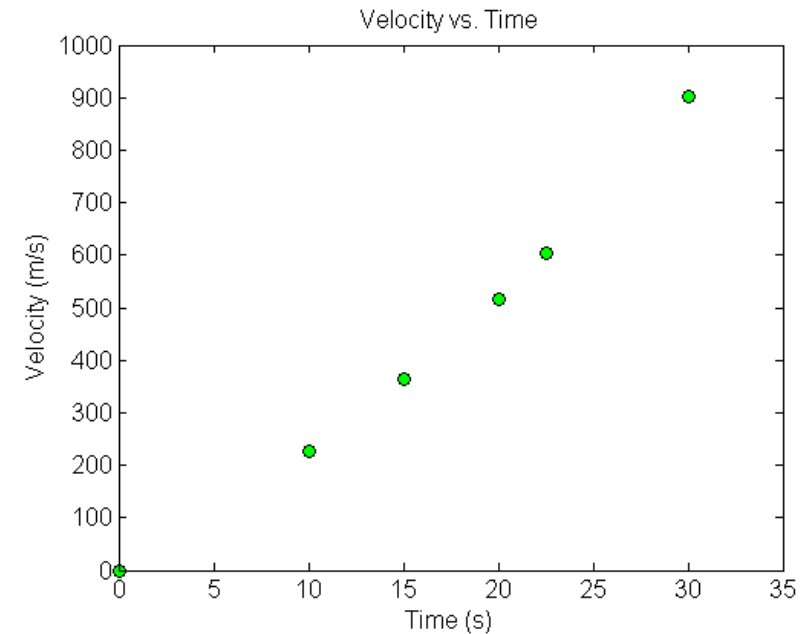


Figure: Velocity vs. time data for the rocket example

Linear Interpolation

$$v(t) = a_0 + a_1 t$$

$$v(15) = a_0 + a_1(15) = 362.78$$

$$v(20) = a_0 + a_1(20) = 517.35$$

Solving the above two equations gives,

$$a_0 = -100.93 \quad a_1 = 30.914$$

Hence

$$v(t) = -100.93 + 30.914t, \quad 15 \leq t \leq 20.$$

$$v(16) = -100.93 + 30.914(16) = 393.7 \text{ m/s}$$

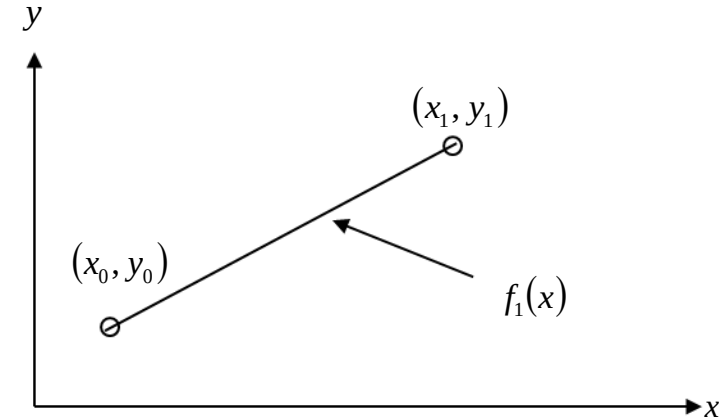


Figure: Linear interpolation.

Example 2

The upward velocity of a rocket is given as a function of time in Table 2.

Find the velocity at $t=16$ seconds using the direct method for quadratic interpolation.

Table 2: Velocity as a function of time.

$t, (s)$	$v(t), (m/s)$
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

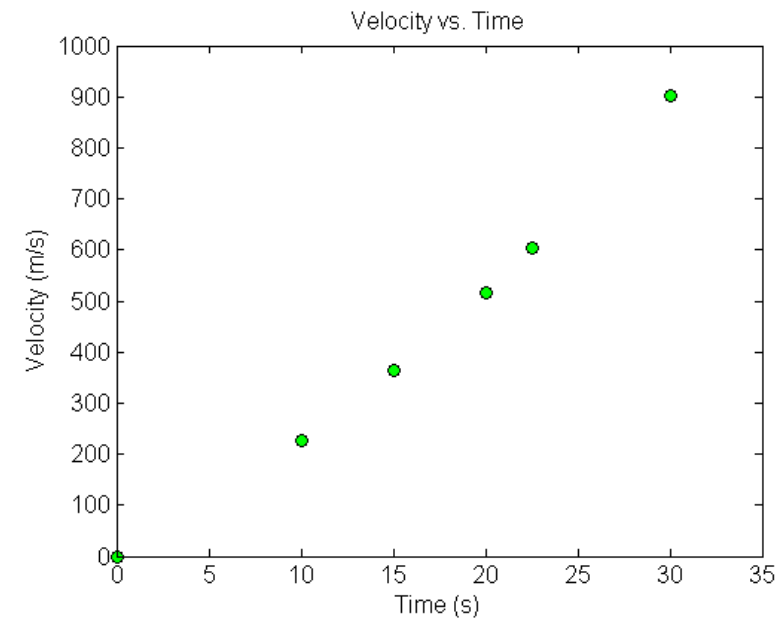


Figure: Velocity vs. time data for the rocket example

Quadratic Interpolation

$$v(t) = a_0 + a_1 t + a_2 t^2$$

$$v(10) = a_0 + a_1(10) + a_2(10)^2 = 227.04$$

$$v(15) = a_0 + a_1(15) + a_2(15)^2 = 362.78$$

$$v(20) = a_0 + a_1(20) + a_2(20)^2 = 517.35$$

Solving the above three equations gives

$$a_0 = 12.05 \quad a_1 = 17.733 \quad a_2 = 0.3766$$

$$v(t) = 12.05 + 17.733t + 0.3766t^2, \quad 10 \leq t \leq 20$$

$$v(16) = 12.05 + 17.733(16) + 0.3766(16)^2 = 392.19 \text{ m/s}$$

The absolute, relative approximate error $|\epsilon_a|$ obtained between the results from the first and second order polynomial is

$$\begin{aligned} |\epsilon_a| &= \left| \frac{392.19 - 393.70}{392.19} \right| \times 100 \\ &= 0.38410\% \end{aligned}$$

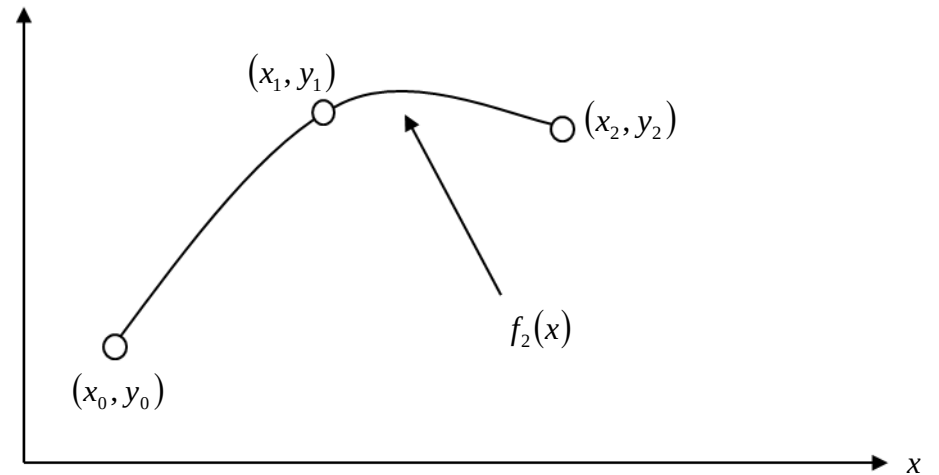


Figure: Quadratic interpolation.

Example 3

The upward velocity of a rocket is given as a function of time in Table 3.

Find the velocity at $t=16$ seconds using the direct method for cubic interpolation.

Table 3 Velocity as a function of time.

$t, (s)$	$v(t), (m/s)$
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

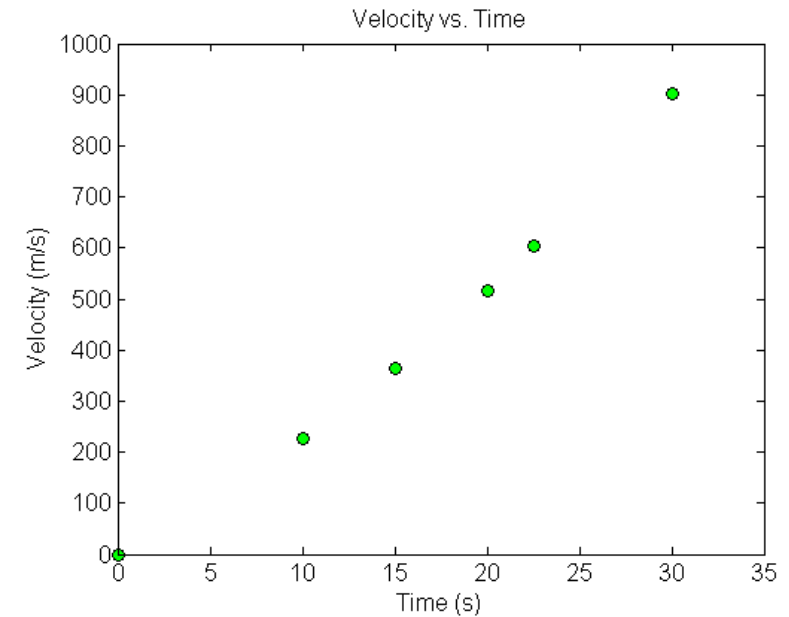


Figure: Velocity vs. time data for the rocket example

Cubic Interpolation

$$v(t) = a_0 + a_1t + a_2t^2 + a_3t^3$$

$$v(10) = 227.04 = a_0 + a_1(10) + a_2(10)^2 + a_3(10)^3$$

$$v(15) = 362.78 = a_0 + a_1(15) + a_2(15)^2 + a_3(15)^3$$

$$v(20) = 517.35 = a_0 + a_1(20) + a_2(20)^2 + a_3(20)^3$$

$$v(22.5) = 602.97 = a_0 + a_1(22.5) + a_2(22.5)^2 + a_3(22.5)^3$$

$$a_0 = -4.2540 \quad a_1 = 21.266 \quad a_2 = 0.13204 \quad a_3 = 0.0054347$$

$$v(t) = -4.2540 + 21.266t + 0.13204t^2 + 0.0054347t^3, \quad 10 \leq t \leq 22.5$$

$$\begin{aligned} v(16) &= -4.2540 + 21.266(16) + 0.13204(16)^2 + 0.0054347(16)^3 \\ &= 392.06 \text{ m/s} \end{aligned}$$

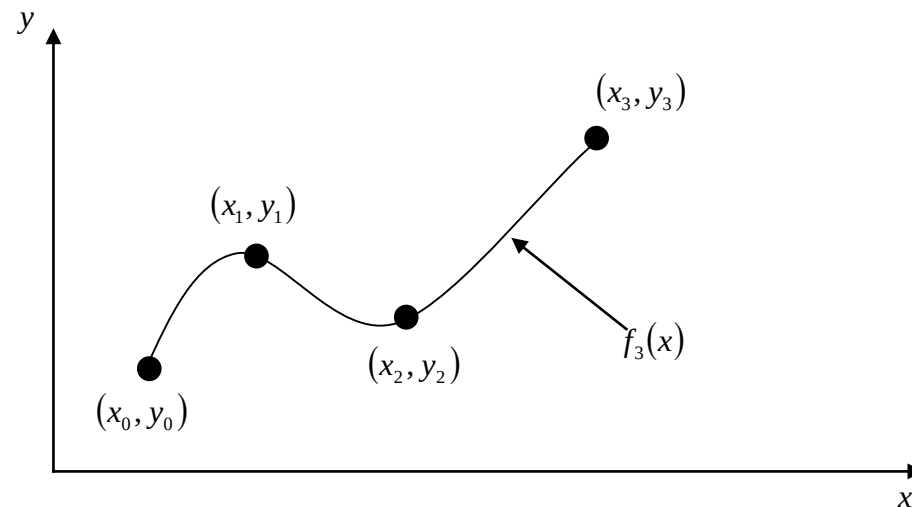


Figure: Cubic interpolation.

The absolute percentage relative approximate error $|\epsilon_a|$ between the second and third order polynomials is

$$\begin{aligned} |\epsilon_a| &= \left| \frac{392.06 - 392.19}{392.06} \right| \times 100 \\ &= 0.033269\% \end{aligned}$$

Comparison Table

t(s)	v (m/s)
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

Table 4: Comparison of different orders of the polynomial.

Order of Polynomial	1	2	3
$v(t = 16) \text{ m/s}$	393.7	392.19	392.06
Absolute Relative Approximate Error	-----	0.38410 %	0.033269 %

LAGRANGE INTERPOLATION

The Lagrange Interpolation Formula finds a polynomial called the Lagrange Polynomial that **takes on certain values at an arbitrary point**. It is an **n th-degree polynomial expression of the function $f(x)$** . The interpolation method is used to find the new data points **within the range of a discrete set** of known data points.

Lagrange Interpolation is a **method for determining the value of a function** at any given point when **the function is not explicitly given**. We use other points on the function to get the value of the function at any required point.

Suppose we have a function $y = f(x)$ in which substituting the values of x gives different values of y .

We are given two points (x_1, y_1) and (x_2, y_2) on the curve, then the value of y at $x = a(\text{constant})$ is calculated using the Lagrange Interpolation Formula.

LAGRANGE INTERPOLATION

Lagrange interpolation polynomial is given by;

$$f_n(x) = \sum_{i=0}^n L_i(x) f(x_i) \quad (1)$$

Where, n in $f_n(x)$ stands for n_{th} order polynomial that approximate the function $y = f(x)$ for $n+1$ number of data points, $(x_0, y_0), (x_1, y_1) \dots (x_n, y_n)$.

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} \quad (2)$$

$L_i(x)$ is a weighting function that includes a product of $n-1$ terms omitting $j=i$ term.

LAGRANGE INTERPOLATION

Now, we get the Lagrange Interpolation Formula as follows,

$$f(x) = \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} \times y_0 + \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} \times y_1 + \dots + \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} \times y_n$$

LAGRANGE INTERPOLATION

Linear interpolation (2 points) is interpolation by the straight line through (x_0, f_0) , (x_1, f_1) ; Thus the linear Lagrange polynomial p_1 is a sum $p_1 = L_0 f_0 + L_1 f_1$, Obviously,

$$L_0(x) = \frac{x - x_1}{x_0 - x_1}, \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}.$$

This gives the linear Lagrange polynomial

$$p_1(x) = L_0(x)f_0 + L_1(x)f_1 = \frac{x - x_1}{x_0 - x_1} \cdot f_0 + \frac{x - x_0}{x_1 - x_0} \cdot f_1. \quad (3)$$

continued

QUADRATIC INTERPOLATION

The interpolation of 3 points given (x_0, f_0) , (x_1, f_1) , (x_2, f_2) by a second-degree polynomial $p_2(x)$, which by Lagrange's idea is

$$p_2(x) = L_0(x)f_0 + L_1(x)f_1 + L_2(x)f_2 \quad (4a)$$

with $L_0(x_0) = 1$, $L_1(x_1) = 1$, $L_2(x_2) = 1$, and $L_0(x_1) = L_0(x_2) = 0$, etc. We claim that

$$\begin{aligned} L_0(x) &= \frac{l_0(x)}{l_0(x_0)} = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} \\ L_1(x) &= \frac{l_1(x)}{l_1(x_1)} = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} \\ L_2(x) &= \frac{l_2(x)}{l_2(x_2)} = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}. \end{aligned} \quad (4b)$$

continued

EXAMPLE 1

Given $(9.0, 2.1972)$, $(9.5, 2.2531)$ and $(11.0, 2.3979)$ are 3 data points. Compute the value of y at $x= 9.2$.

Solution. 3 points make a second-degree polynomial $p_2(x)$, which by Lagrange's idea is

$$p_2(x) = L_0(x)f_0 + L_1(x)f_1 + L_2(x)f_2$$

Where,
$$L_0(x) = \frac{(x - 9.5)(x - 11.0)}{(9.0 - 9.5)(9.0 - 11.0)} = x^2 - 20.5x + 104.5, \quad L_0(9.2) = 0.5400,$$

$$L_1(x) = \frac{(x - 9.0)(x - 11.0)}{(9.5 - 9.0)(9.5 - 11.0)} = -\frac{1}{0.75} (x^2 - 20x + 99), \quad L_1(9.2) = 0.4800,$$

$$L_2(x) = \frac{(x - 9.0)(x - 9.5)}{(11.0 - 9.0)(11.0 - 9.5)} = \frac{1}{3} (x^2 - 18.5x + 85.5), \quad L_2(9.2) = -0.0200,$$

Putting the values of $L(x)$ in the above equation we get ,

$$p_2(9.2) = 0.5400 \cdot 2.1972 + 0.4800 \cdot 2.2513 - 0.0200 \cdot 2.3979 = 2.2192.$$

Example 2

Determine the value of $f(x)$ or y at $x=23$ from the following table using Lagrange interpolation method.

x	3	6	11	14	25
y or f	1	10	171	93	15

[Remember 5 points produce 4th order polynomial

$$p_4(x) = L_0(x) f_0 + L_1(x) f_1 + L_2(x) f_2 + L_3(x) f_3 + L_4(x) f_4$$

$$\begin{aligned}\text{Where, } L_0(23) &= \frac{(23-6)(23-11)(23-14)(23-25)}{(3-6)(3-11)(3-14)(3-25)} = \frac{(17)(12)(9)(-2)}{(-1)(-8)(-11)(-22)} \\ &= \frac{-3672}{1936} = -1.897\end{aligned}$$

At the same way, calculate $L_1(23)$ and so on

LAGRANGE INTERPOLATION

Properties of Lagrange Interpolation Formula

- This formula is used to find the value of the function at any point, even when the function itself is not given.
- It is used even if the points given are not evenly spaced.
- It gives the value of the dependent variable for any independent variable belonging to any function and thus is used in Numerical Analysis for finding the values of the function, etc.
- It is used in image scaling, AI modeling, NLPs, etc.